

A Muslim School Advantage?

Evidence from a Natural Experiment

Said Hassan

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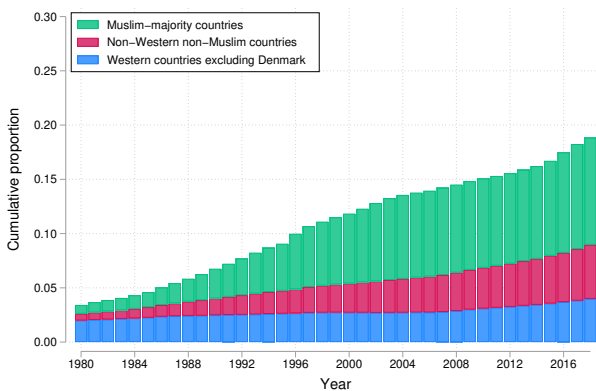


Motivation

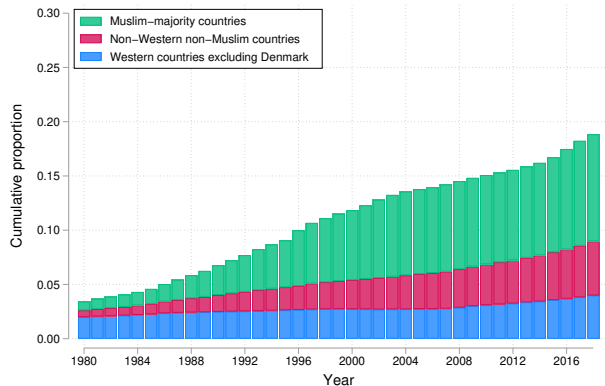
Do Muslim schools improve the academic performance of Muslim children?

Motivation

Figure: School-aged children (5–15 yrs) by ethnic background in Denmark.

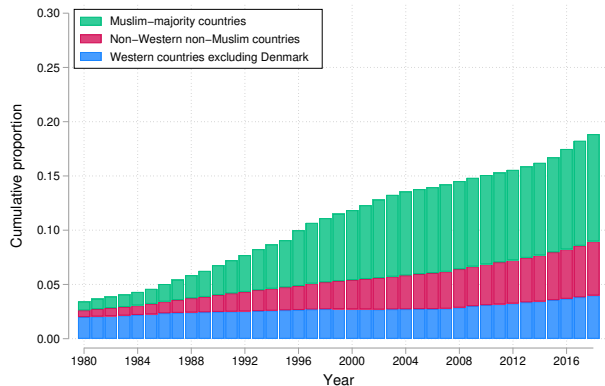


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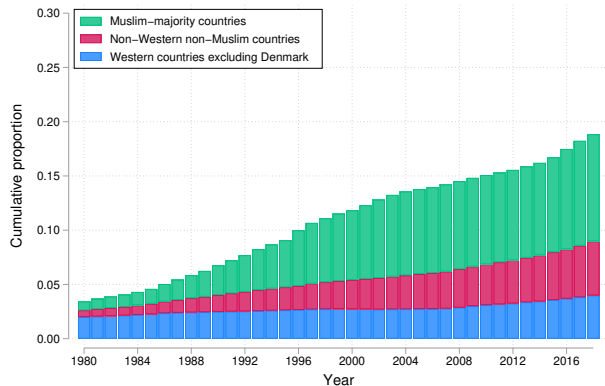
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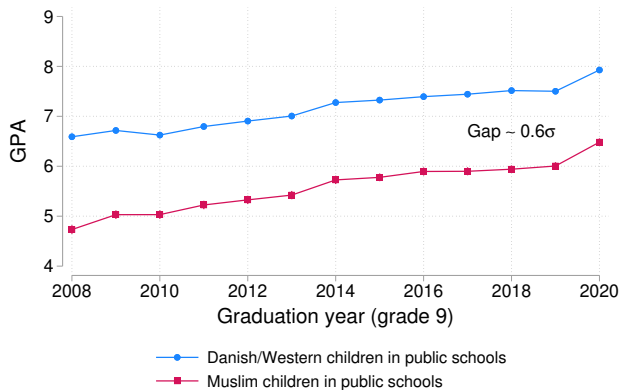
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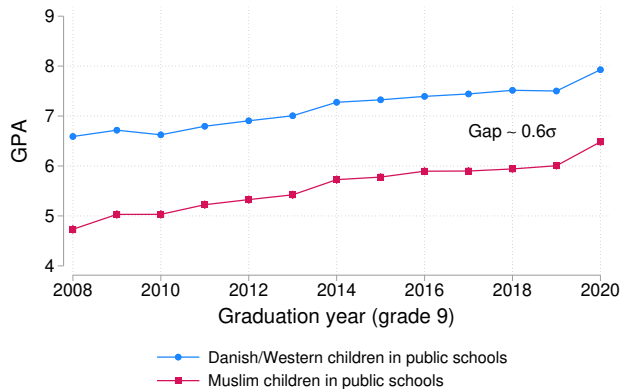
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- Socioeconomically disadvantaged (vs natives/Western immigrants)
(Heath et al., 2008; Alba et al., 2011; Dustmann et al., 2017)
- Concerns regarding “parallel societies” and social integration
(Heath and Demireva, 2014; Hilbig and Riaz, 2022)

Motivation

Figure: Test score gap: Muslim vs. non-Muslim students in **public schools**

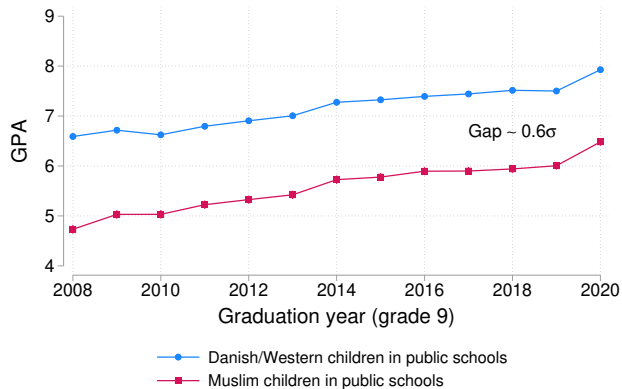


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➤ $\approx$ Black-White achievement gap in US (Reardon, 2015)

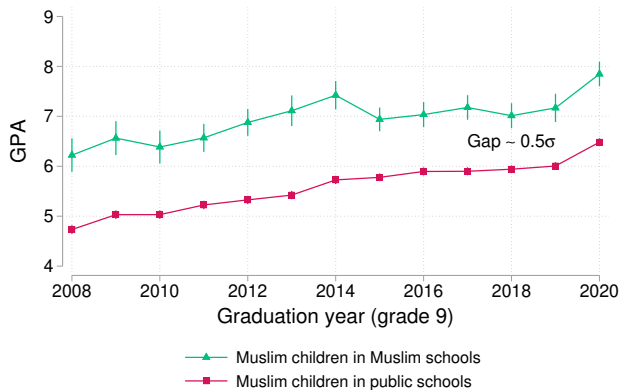
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- $\approx$ Black-White achievement gap in US (Reardon, 2015)
- Muslim schools $\Rightarrow$ negative peer effects? (Sacerdote, 2011)

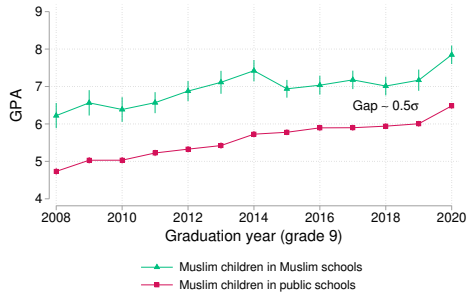
Motivation

Figure: Performance of **Muslim kids** in public vs Muslim schools



Motivation

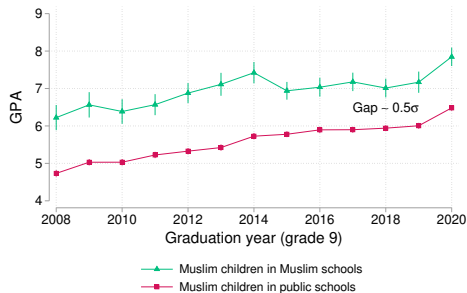
Selection? ... or positive effect?



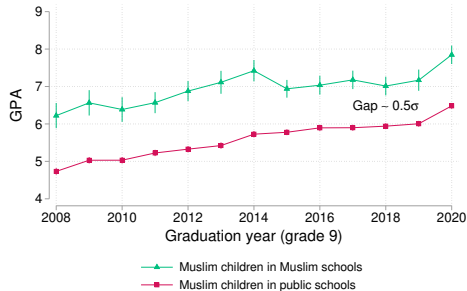
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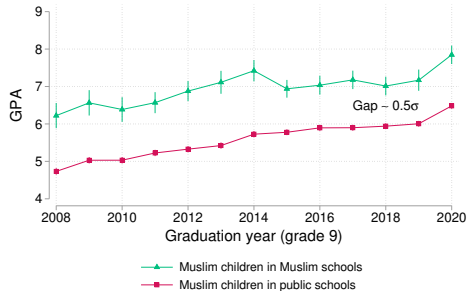
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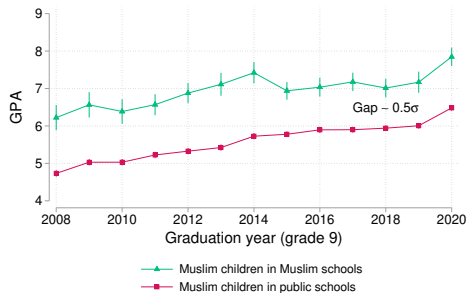
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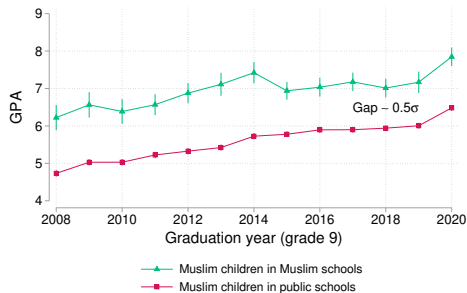
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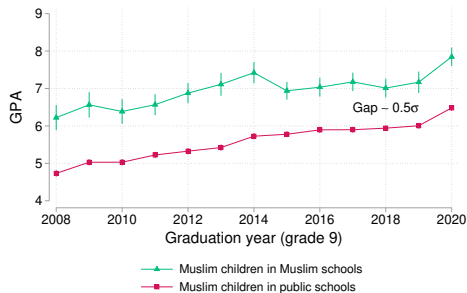
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- Discrimination/teacher bias (Bates et al., 2013)

Roadmap: Remainder of this talk

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5. Long term outcomes and social integration

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Danish context

- 150 year tradition of independent schools
(uptake $\approx$ 20% in 2018)

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▶ Compare muslim schools

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- Precise educational records/history

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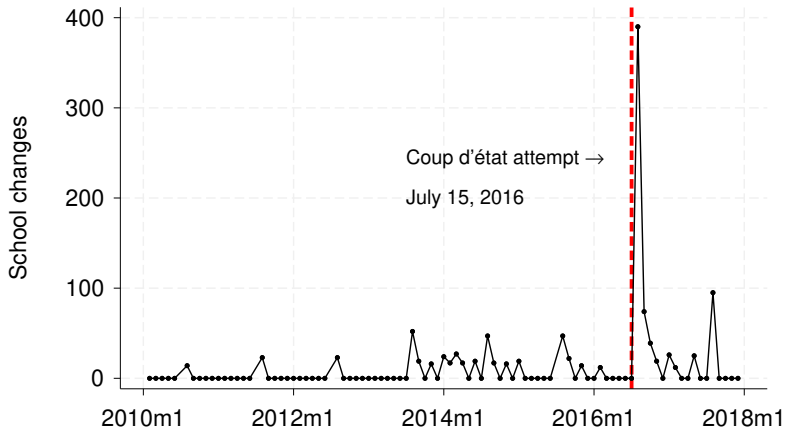
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Full population register data

- Precise educational records/history
- Parental origin country
- Test scores in grade 9 (but no panel data!)
- Reading and mathematics
 - Main: written externally evaluated
 - Supplementary: teacher evaluated

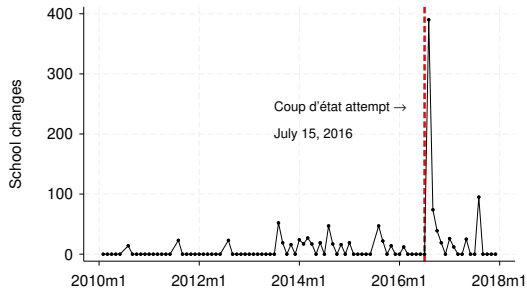
Natural experiment

Gülen → Public school



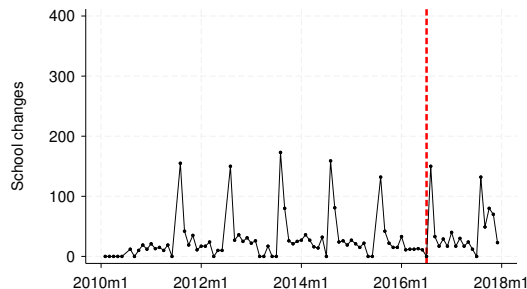
Natural experiment

Gülen → Public school



► Covariate balance

Muslim (non-Gülen) → Public



Main findings

	Reading			Math		
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	OLS	IPTW	OLS	OLS	IPTW
Muslim school	0.345** (0.115)	0.319** (0.103)	0.322** (0.123)	0.319** (0.110)	0.283** (0.094)	0.319** (0.112)
Constant	-0.250 (0.158)	-0.202 (0.151)	-0.520*** (0.116)	-0.388*** (0.107)	-0.137 (0.119)	-0.694*** (0.096)
<i>Observations</i>	660	660	660	644	644	644
<i>R</i> ²	0.056	0.175		0.060	0.137	
Covariates	No	Yes	Yes [†]	No	Yes	Yes [†]

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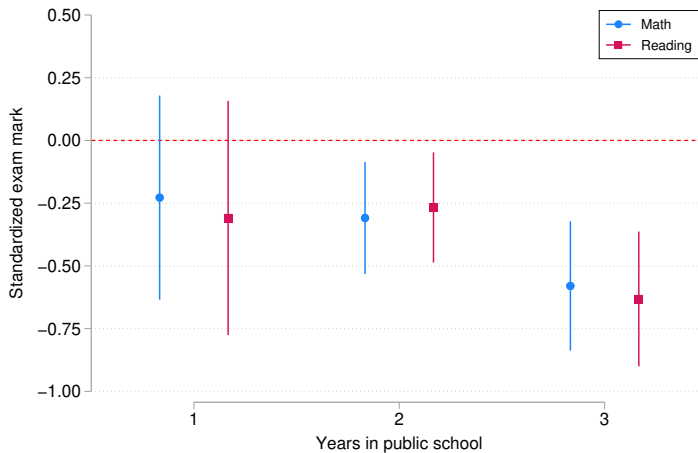
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- 1-year school learning estimates range from $1/4$ – $1/3 \sigma$ (Woessmann, 2016)
- **How much is driven by disruption?**

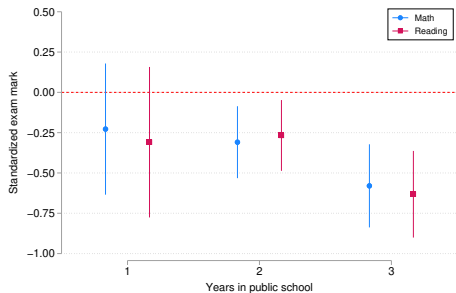
Disruption effects

Variation in exposure to public schools



Disruption effects

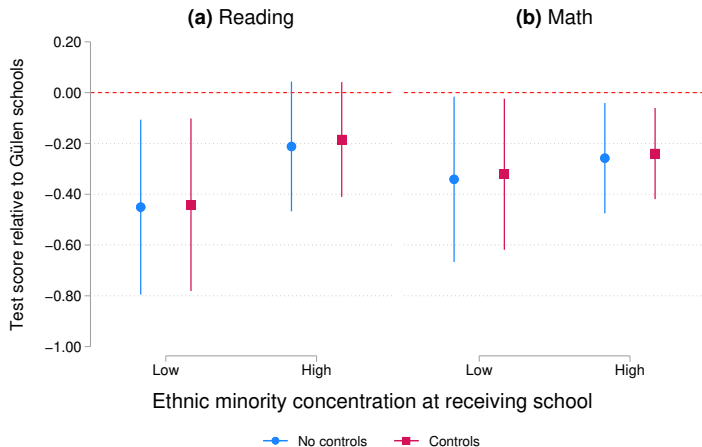
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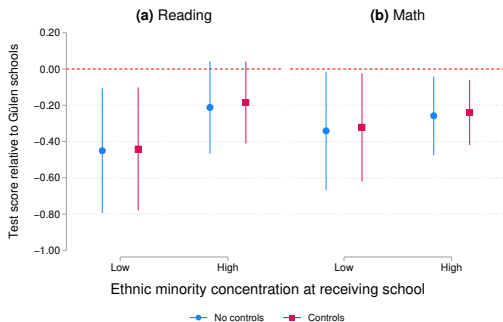
Supplementary analyses:

1. Benchmarking disruption ($\approx 0.08\sigma$)
2. Transitions between school types
3. Changes to non-Gülen Muslim schools
4. Event study: parental stress
5. Event study: student performance

Mechanism I: Ethnic similarity



Mechanism I: Ethnic similarity



➤ Robust to alternative specifications

▸ Ethnicity definitions

▸ Percentile cutoffs

➤ **Not** driven by SES

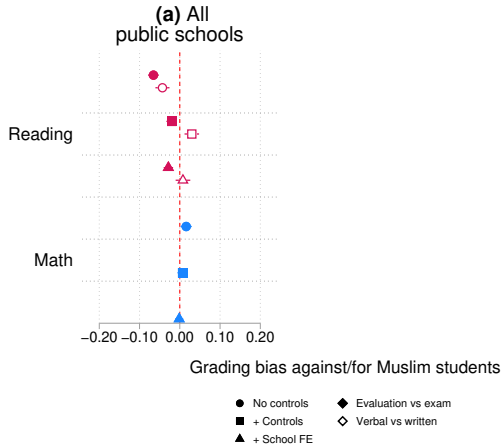
▸ Control for SES

▸ Descriptives

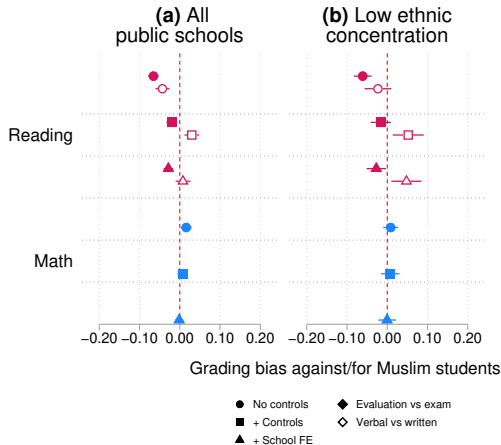
➤ Aspirations play a role

▸ Aspiration results

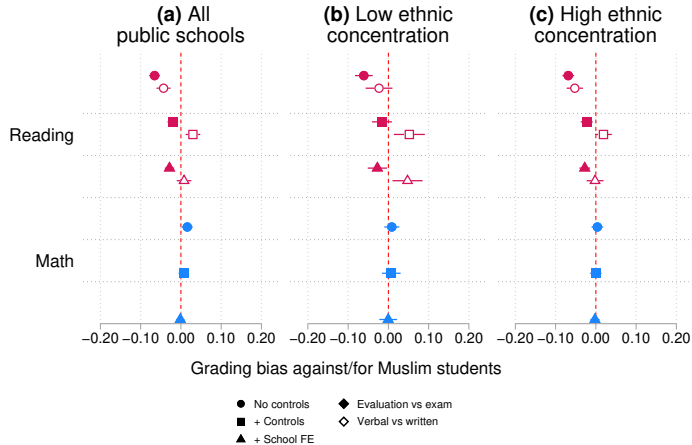
Mechanism II: Grading bias and discrimination



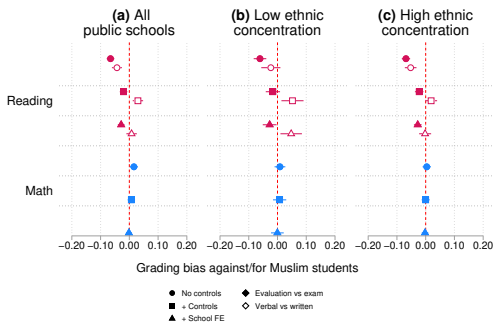
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➤ Robust to different models, samples, measures

▸ Conditional vs difference

▸ Comparison groups

▸ Sample periods

➤ Subtle biases could still persist

▸ Subjective evaluations

Conclusion

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- Not explained by grading bias
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Policy

- Assimilationist policies come with costs to important outcomes
- Limitation: no data on values, norms, etc.

Thank You!

saidhassan.net

Long-term outcomes and integration

	High school outcomes			Social benefits recipient	
	Admission (age 18)	Completion (age 20)	Standardized GPA	Ever	Weeks
	(1)	(2)	(3)	(4)	(5)
Muslim school	0.113** (0.037)	0.122* (0.058)	0.355* (0.149)	−0.048 (0.033)	−6.397 (4.828)
Constant	−0.172*** (0.048)	−0.163 (0.082)	−1.175*** (0.185)	0.029 (0.038)	3.024 (3.930)
<i>Observations</i>	720	720	514	720	720
<i>R</i> ² adj.	0.078	0.118	0.125	0.041	0.019
Covariates	Yes	Yes	Yes	Yes	Yes

Covariate balance

	Treatment		Control		Difference	<i>p</i> -value
	Mean	(SD)	Mean	(SD)		
Female	0.563	(0.496)	0.488	(0.501)	0.075	0.086
Age	16.153	(0.653)	16.226	(0.501)	−0.073	0.182
Both parents born in DK	0.815	(0.388)	0.774	(0.420)	0.041	0.236
Birth weight (kg)	3.412	(0.548)	3.413	(0.645)	−0.001	0.978
Parents divorced	0.080	(0.271)	0.071	(0.258)	0.008	0.726
Parental education (years)	12.823	(2.605)	12.644	(2.549)	0.179	0.436
Parental income	71.059	(30.889)	73.088	(25.237)	−2.028	0.438
Parent unemployed	0.198	(0.399)	0.263	(0.442)	−0.066	0.070
<i>Observations</i>	552		168			

Comparing movers to stayers

	Stayers		Movers		Difference	<i>p</i> -value
	Mean	(SD)	Mean	(SD)		
Female	0.562	0.497	0.488	0.501	0.074	0.126
Age	16.121	0.748	16.226	0.501	-0.105	0.105
Both parents born in DK	0.776	0.418	0.774	0.420	0.002	0.958
Birth weight (kg)	3.407	0.553	3.413	0.645	-0.006	0.917
Parents divorced	0.094	0.292	0.071	0.258	0.022	0.411
Parental education (years)	12.962	2.629	12.644	2.549	0.318	0.209
Parental income	73.764	33.797	73.088	25.237	0.676	0.821
Parent unemployed	0.182	0.387	0.263	0.442	-0.081	0.040
<i>Observations</i>	299		168			

Disruption: Benchmark VA estimates

[▶ Back to robust](#)

	Full sample			Muslim children		
	(1)	(2)	(3)	(4)	(5)	(6)
Disruption (δ)	−0.104*** (0.010)	−0.096*** (0.009)	−0.080*** (0.008)	−0.080*** (0.022)	−0.068** (0.021)	−0.063*** (0.019)
α	0.683*** (0.003)	0.615*** (0.003)	0.601*** (0.003)	0.544*** (0.006)	0.504*** (0.006)	0.493*** (0.006)
γ			0.381*** (0.008)			0.338*** (0.018)
Constant	0.001 (0.004)	1.721*** (0.057)	1.791*** (0.057)	−0.183*** (0.010)	2.086*** (0.149)	2.221*** (0.150)
Observations	373,576	373,575	373,575	35,076	35,076	35,076
R^2	0.431	0.496	0.510	0.365	0.421	0.437
Covariates	No	Yes	Yes	No	Yes	Yes

Disruption: Transition analyses [▶ Back to robust](#)

	From public school			From private school		From Muslim school	
	To public (1)	To private (2)	To Muslim (3)	To private (4)	To public (5)	To Muslim (6)	To public (7)
Move	−0.166*** (0.011)	0.039 (0.027)	0.163* (0.077)	−0.189*** (0.028)	−0.368*** (0.021)	0.043 (0.123)	−0.489*** (0.086)
Constant	−0.166*** (0.007)	−0.182*** (0.006)	−0.180*** (0.006)	0.101*** (0.016)	0.080*** (0.016)	−0.163 (0.124)	−0.153 (0.112)
<i>Obs. (total)</i>	256,697	265,828	265,828	43,223	47,647	1,566	1,837
<i>Obs. (movers)</i>	33,827	9,072	113	2,132	4,476	45	269
R^2 adj.	0.189	0.185	0.185	0.142	0.158	0.064	0.113
Covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Disruption: Gülen → other Muslim schools

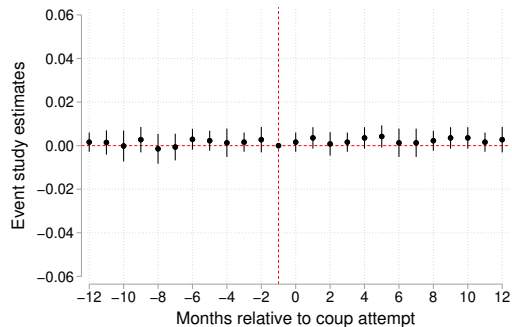
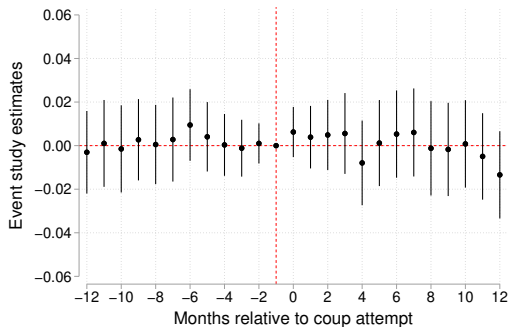
	Reading		Math	
	(1)	(2)	(3)	(4)
Movers	−0.184 [−0.525,0.156] (0.264)	−0.018 [−0.348,0.312] (0.910)	−0.088 [−0.469,0.292] (0.624)	0.065 [−0.357,0.488] (0.743)
Constant	0.095 [−0.211,0.402] (0.514)	0.108 [−0.210,0.427] (0.476)	−0.069 [−0.148,0.010] (0.080)	0.124 [−0.009,0.258] (0.066)
<i>Observations</i>	663	663	649	649
R^2	0.022	0.144	0.015	0.089
Covariates	No	Yes	No	Yes

Disruption: Event studies of parental reactions

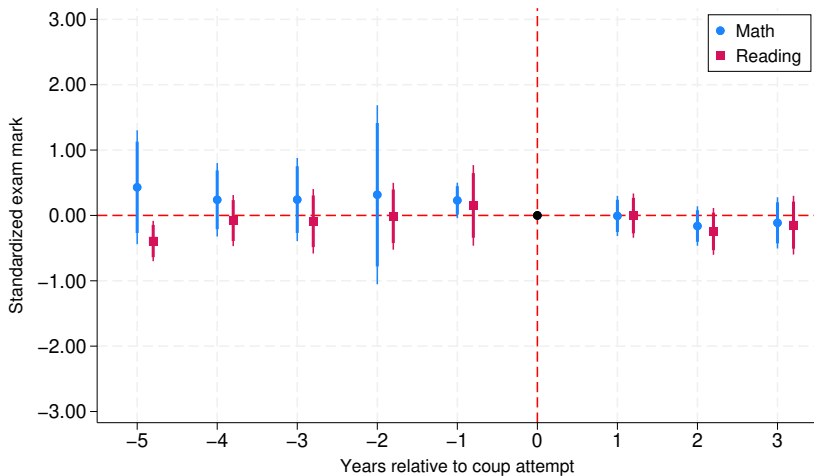
Figure: Event study: parents with children in Gülen vs non-Gülen Turkish Muslim schools

(a) Sick leave

(b) Divorce



Disruption: Time trends



Mechanism I: Ethnic similarity – School and teacher characteristics [▶ Back](#)

	Low Ethnic Concentration		High Ethnic Concentration		<i>t</i> -test	
	Mean	(SD)	Mean	(SD)	Difference	<i>p</i> -value
<i>School and peer characteristics</i>						
Ethnic minority	0.274	(0.094)	0.603	(0.174)	−0.329	0.000
School GPA (std)	−0.130	(0.267)	−0.347	(0.248)	0.217	0.000
School cohort size	188.013	(50.768)	167.988	(56.120)	20.025	0.019
Paternal income	59.919	(9.740)	47.384	(6.937)	12.535	0.000
Maternal income	47.152	(4.703)	38.164	(4.241)	8.988	0.000
Paternal education	13.969	(0.920)	13.020	(0.500)	0.949	0.000
Maternal education	13.793	(1.003)	12.495	(0.910)	1.298	0.000
<i>Teacher characteristics</i>						
Ethnic minority	0.053	(0.050)	0.115	(0.076)	−0.062	0.000
Age	42.541	(2.716)	42.289	(1.702)	0.252	0.481
Certified	0.801	(0.093)	0.795	(0.087)	0.006	0.677
Experience (years)	12.678	(2.350)	12.566	(1.735)	0.111	0.733
Experience ≥ 3 years	0.887	(0.064)	0.879	(0.057)	0.008	0.390
<i>Observations</i>	77		83			

Ethnic similarity - Alternative definitions [▶ Back](#)

	A. Main results: Any minority		B. Alternative measure: Muslim		C. Alternative measure: Turkish	
	Reading (1)	Math (2)	Reading (3)	Math (4)	Reading (5)	Math (6)
<i>Ethnic concentration</i>						
Low (below median)	−0.441*	−0.321*	−0.397*	−0.278	−0.452***	−0.339***
	(0.171)	(0.150)	(0.165)	(0.144)	(0.118)	(0.099)
High (above median)	−0.185	−0.240**	−0.224	−0.280**	−0.206	−0.236
	(0.114)	(0.090)	(0.117)	(0.088)	(0.146)	(0.123)
<i>Observations</i>	655	639	655	639	655	639
<i>R²</i>	0.168	0.128	0.166	0.127	0.168	0.128
<i>Covariates</i>	Yes	Yes	Yes	Yes	Yes	Yes

Ethnic similarity - Alternative pctile cutoffs [▶ Back](#)

	Reading			Math		
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Ethnic concentration</i>						
Low ($< P_{25}$)	−0.509*** (0.128)			−0.321 (0.163)		
High ($\geq P_{25}$)	−0.269* (0.119)			−0.271** (0.101)		
Low ($< P_{50}$)		−0.445* (0.175)			−0.329* (0.153)	
High ($\geq P_{50}$)		−0.187 (0.112)			−0.235* (0.089)	
Low ($< P_{75}$)			−0.334** (0.122)			−0.296** (0.104)
High ($\geq P_{75}$)			−0.221 (0.190)			−0.222 (0.139)
Observations	655	655	655	639	639	639
R^2	0.166	0.168	0.164	0.127	0.128	0.127
Covariates	Yes	Yes	Yes	Yes	Yes	Yes

Ethnic similarity - Correlation between measures [▶ Back](#)

Variables	A. All Schools			B. Receiving Schools		
	1.	2.	3.	1.	2.	3.
1. % Muslim	1.000			1.000		
2. % Any non-western	0.962	1.000		0.980	1.000	
3. % Turkish	0.631	0.586	1.000	0.645	0.599	1.000

Ethnic similarity - Control for SES [▶ Back](#)

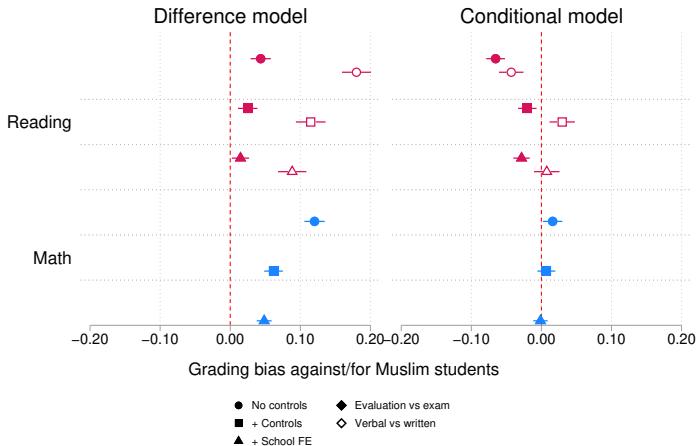
	Reading				Math			
	Main				Main			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Ethnic concentration</i>								
Low	-0.441*	-0.444	-0.331	-0.459*	-0.321*	-0.573	-0.288	-0.329*
	(0.171)	(0.250)	(0.389)	(0.179)	(0.150)	(0.298)	(0.379)	(0.163)
High	-0.185	-0.185	-0.117	-0.192	-0.240**	-0.335**	-0.220	-0.193*
	(0.114)	(0.125)	(0.223)	(0.163)	(0.090)	(0.125)	(0.220)	(0.092)
<i>School variables (average in $t - 1$)</i>								
Parental income		0.000				0.000		
		(0.000)				(0.000)		
Parental education			-0.049				-0.015	
			(0.126)				(0.129)	
School GPA				0.020				0.143
				(0.323)				(0.203)
Observations	655	655	655	637	639	639	639	621

Ethnic similarity - Aspirations

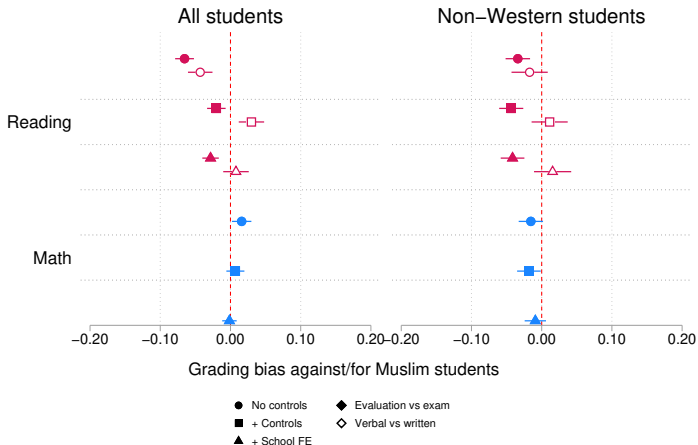
[▶ Back](#)

	<i>Dep. var.: Educational aspirations</i>			
	(1)	(2)	(3)	(4)
Minority student	0.006 (0.012)	0.097*** (0.018)	0.108*** (0.022)	0.075*** (0.020)
% Minority in school			0.000 (0.000)	
Minority student × % Minority in school			−0.001 (0.001)	0.000 (0.000)
Constant	0.794*** (0.005)	0.748*** (0.005)	0.741*** (0.007)	0.749*** (0.003)
<i>Observations</i>	46,988	44,105	36,212	36,209
<i>R</i> ² adj.	0.000	0.197	0.198	0.251
Covariates	No	Yes	Yes	Yes
School fixed effects	No	No	No	Yes

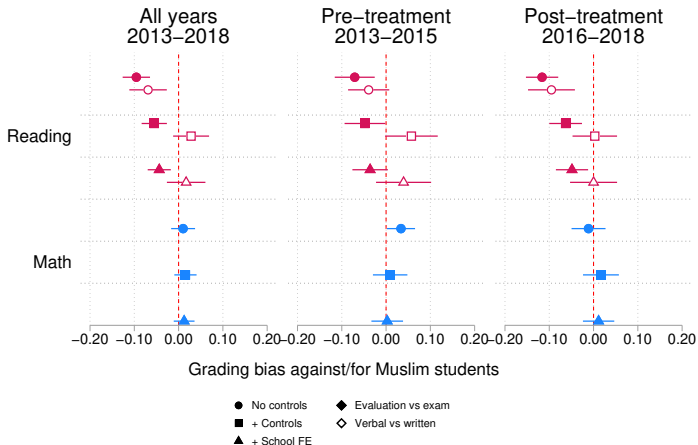
Grading bias: Conditional vs difference models

[▶ Back](#)

Grading bias: Comparison groups

[▶ Back](#)

Grading bias: Sample periods (receiving schools)

[▶ Back](#)

Subjective evaluations

[▶ Back](#)

	<i>Dep. var.: Teacher-evaluated student readiness for high school</i>					
	Academic readiness		Social readiness		Personal readiness	
	(1)	(2)	(3)	(4)	(5)	(6)
Muslim student	−0.154*** (0.008)	−0.030*** (0.008)	−0.133*** (0.009)	−0.041*** (0.009)	−0.181*** (0.009)	−0.057*** (0.009)
Constant	0.829*** (0.003)	0.784*** (0.003)	0.892*** (0.003)	0.832*** (0.004)	0.831*** (0.003)	0.746*** (0.005)
<i>Observations</i>	94,992	94,992	73,830	73,830	73,830	73,830
<i>R² adj.</i>	0.012	0.240	0.016	0.106	0.020	0.149
Covariates	No	Yes	No	Yes	No	Yes

Robustness: School closures only

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[▶ Back to robust](#)

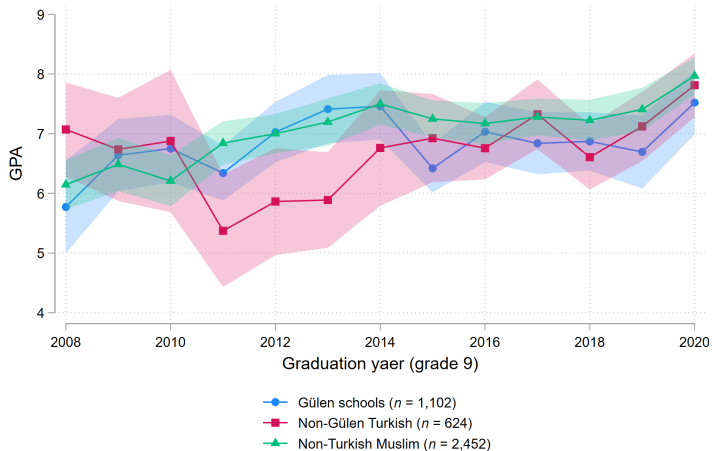
	Reading			Math		
	(1) OLS	(2) OLS	(3) IPTW	(4) OLS	(5) OLS	(6) IPTW
Muslim school	0.458 (0.285)	0.601* (0.280)	0.226 (0.359)	0.379 (0.272)	0.657* (0.254)	0.123 (0.185)
Constant	-0.226 (0.285)	-0.277 (0.330)	-0.587*** (0.113)	-0.517 (0.272)	-0.529* (0.260)	-0.671*** (0.103)
<i>Observations</i>	148	148	148	148	148	148
<i>R</i> ²	0.105	0.259		0.064	0.201	
Covariates	No	Yes	Yes [†]	No	Yes	Yes [†]

Robustness: Results without stayers

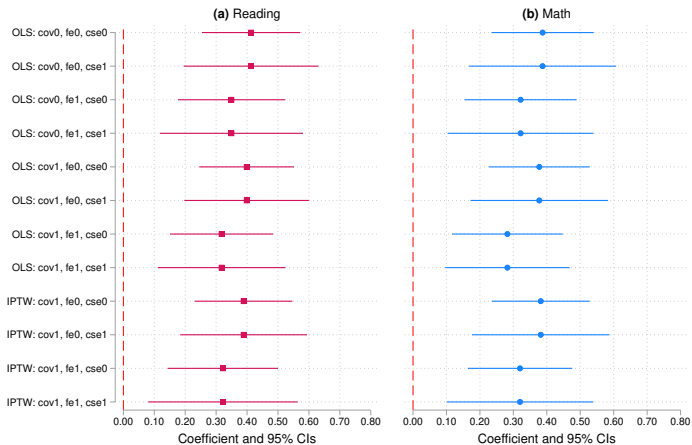
[▶ Back to main](#)
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	Reading			Math		
	(1)	(2)	(3)	(4)	(5)	(6)
	OLS	OLS	IPTW	OLS	OLS	IPTW
Muslim school	0.884*** (0.184)	0.923*** (0.181)	0.469*** (0.123)	0.868*** (0.099)	0.858*** (0.143)	0.465*** (0.119)
Constant	−0.788*** (0.122)	−0.700*** (0.185)	−0.587*** (0.086)	−0.937*** (0.093)	−0.688*** (0.179)	−0.756*** (0.073)
<i>Observations</i>	379	379	379	379	379	379
<i>R</i> ²	0.089	0.202		0.090	0.150	
Covariates	No	Yes	Yes [†]	No	Yes	Yes [†]

Robustness: Gülen vs other Muslim schools - test scores

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Alternative specifications



Unemployment variables

Table: Summary statistics using different employment variables by treatment status.

Parental employment status	Treatment		Control		<i>t</i> -test	
	Mean	(SD)	Mean	(SD)	Diff	<i>p</i> -value
Employed	0.817	(0.387)	0.778	(0.417)	0.039	0.269
Unemployed	0.198	(0.399)	0.263	(0.442)	−0.066	0.070
Not in labor force	0.497	(0.500)	0.443	(0.498)	0.054	0.222
Unemployed/not in labor force	0.623	(0.485)	0.641	(0.481)	−0.018	0.678
<i>Observations</i>	552		168			